

Questions & Answers

Chapter 2 — Cybersecurity

Programming and Artificial Intelligence • Egyptian Baccalaureate • 2nd Year

2-1	Cryptographic Technologies and Authentication
2-2	Network Security Design
2-3	Incident Response and Risk Management

Question styles: True/False • Fill in the blanks • Matching • Multiple Choice • Short Answer • 6-Mark •
Think as an Engineer • New Context • Reflect & Challenge

2-1 — Cryptographic Technologies and Authentication

1. Mark each statement (A–D) with ■ if true or × if false.

Question

A. HTTPS combines public-key cryptography and common-key cryptography. (■ True / × False)

Answer: True — Public-key cryptography is used for safe key sharing; common-key cryptography is used for data communication.

Question

B. HTTPS uses public-key cryptography to exchange all data after the common key is shared. (■ True / × False)

Answer: False — Common-key cryptography is used for data communication because it is faster.

Question

C. Multi-factor authentication combines two or more different authentication factors. (■ True / × False)

Answer: True — The factors are knowledge, possession, and biometric.

Question

D. Using two different passwords is an example of multi-factor authentication. (■ True / × False)

Answer: False — Both passwords are knowledge factors.

2. Answer the following questions.

Question

1. In HTTPS, which cryptographic method is used so the browser and server can safely share a common key?

Answer: Public-key cryptography.

Question

2. Which cryptographic method is used for data communication after the common key is shared?

Answer: Common-key cryptography (symmetric-key cryptography).

Question

3. Why are the two cryptographic methods combined?

Answer: Public-key cryptography shares the key safely, while common-key cryptography exchanges large amounts of data quickly. This provides both security and speed.

3. Matching — security technology and threat

Question

Match each description with the most appropriate choice.

Choices: A Encryption • B Digital signature • C Digital certificate • D Multi-factor authentication

a Prevent communication content from being read by third parties.

b Detect whether data has been altered during communication.

c Verify that the communication partner is authentic.

d Prevent unauthorized login even if a password is leaked.

Answer: a → A • b → B • c → C • d → D

4. Matching — authentication factors

Question

Match each description with the most appropriate choice.

Choices: A Knowledge + possession • B Knowledge + biometric • C Biometric + possession

a Password + one-time password

b Password + fingerprint authentication

c Password + SMS authentication

d Fingerprint authentication + smart card

Answer: a → A • b → B • c → A • d → C

5. Multiple Choice

Question

Which is NOT an appropriate description of security technologies?

A Encryption can prevent communication content from being read by third parties.

B A digital signature is mainly used to improve communication speed.

C A digital certificate helps verify that the communication partner is authentic.

D Combining several security technologies can address different threats.

Answer: B — Improving communication speed is not the purpose of a digital signature.

6. Fill in the blanks

Question

The web server sends a (a) and a public key to the browser. The browser verifies the server through (b). Next, the browser creates a (c) for communication and protects it using the server's public key. Data communication then uses (c).

Answer: (a) digital certificate • (b) certificate authority (CA) • (c) common key.

7. Multiple Choice — MFA

Question

Which is NOT an appropriate example of multi-factor authentication?

A Password + smartphone notification

B Password + secret question

C Fingerprint authentication + smart card

D Password + SMS authentication

Answer: B — A password and a secret question are both knowledge factors.

8. Short Answer

Question

Name the three authentication factors.

Answer: Knowledge, possession, and biometric.

Question

What does a digital certificate verify?

Answer: It verifies that the communication partner, such as a website, is authentic.

Question

What does a digital signature help detect?

Answer: Data tampering and impersonation; it also provides non-repudiation.

EXAM-STYLE QUESTION — [6 MARKS]

Question

Explain why HTTPS uses public-key cryptography to share the key but common-key cryptography to exchange the data.

- Answer:**
1. Public-key cryptography is used to share the common key safely.
 2. The browser and server use the public/private key process during key sharing.
 3. After the common key is established, common-key cryptography is used for the data.
 4. Common-key cryptography is faster for large amounts of data.
 5. Using public-key cryptography for all data communication would be inefficient.
 6. Therefore, combining both methods provides security and communication speed.

9. Think as an Engineer — Investigate & Decide

Question

A school portal lets students check grades and pay fees. Which is better: password only or password plus a one-time password sent to the phone? Give evidence.

Answer: Password + one-time password is the stronger choice because it combines knowledge and possession. If the password leaks, the second factor can still prevent unauthorized access. A student without a smartphone may need an alternative.

10. New Context

Question

A small online shop wants to secure customer login and payment. Name two security technologies, say where each helps, and identify one threat that could still remain.

Answer: For example, HTTPS/encryption protects communication content and MFA protects account login. A digital certificate can verify the genuine site. A remaining risk could be a compromised user device or account.

2-2 — Network Security Design

1. Mark each statement (A–D) with ■ if true or × if false.

Question

A. A firewall allows or denies communication based on rules set in advance. (■ True / × False)

Answer: True — This is the role of a firewall at the network entry/exit point.

Question

B. Using a VPN can reduce the risk of eavesdropping on public Wi-Fi. (■ True / × False)

Answer: True — VPN communication is encrypted.

Question

C. A DMZ exposes all servers directly to the outside. (■ True / × False)

Answer: False — A DMZ separates public-facing servers from the internal network.

Question

D. Defense in depth means layering multiple security measures. (■ True / × False)

Answer: True — If one layer is breached, another layer can still protect the system.

2. Answer the following questions.

Question

1. What is the term for a system at a network entry/exit point that allows or denies communication according to rules?

Answer: Firewall.

Question

2. What is the three-letter abbreviation for an encrypted virtual private line over the Internet?

Answer: VPN — Virtual Private Network.

Question

3. What is the three-letter abbreviation for the area where public-facing servers are separated from the internal network?

Answer: DMZ — Demilitarized Zone.

Question

4. What is the concept of layering multiple security measures?

Answer: Defense in depth.

Question

5. What concept means that no communication is trusted automatically and every access is verified?

Answer: Zero trust.

3. Matching

Question

Match each description with the most appropriate choice.

Choices: A Firewall • B VPN • C DMZ • D Defense in depth

a Creates an encrypted virtual private line over the Internet for safe remote connection.

b Places public-facing servers separately from the internal network.

c Allows or denies communication based on preset rules.

d Layers multiple security measures.

Answer: $a \rightarrow B \bullet b \rightarrow C \bullet c \rightarrow A \bullet d \rightarrow D$

4. Multiple Choice — VPN

Question

Which is NOT an appropriate use of a VPN?

A Connecting safely from home to an organization's network.

B Connecting offices in different locations safely.

C Automatically deleting files infected with viruses.

D Reducing eavesdropping risk on public Wi-Fi through encrypted communication.

Answer: **C** — Deleting infected files is not a VPN function.

5. Matching — defense-in-depth layers

Question

Match each description with the most appropriate choice.

Choices: **A** Network entry • **B** Communication path • **C** Server placement • **D** Endpoints

a Blocking unauthorized communication with a firewall.

b Encrypting communication with a VPN.

c Installing antivirus software and keeping the operating system updated.

d Separating public-facing servers using a DMZ.

Answer: $a \rightarrow A \bullet b \rightarrow B \bullet c \rightarrow D \bullet d \rightarrow C$

6. Multiple Choice — DMZ

Question

What is the most appropriate reason for providing a DMZ?

A To improve communication speed.

B To expose all servers to the outside.

C To reduce the chance that an attack on a public-facing server reaches the internal network.

D To automatically detect and remove viruses.

Answer: **C** — A DMZ separates public-facing servers from the internal network.

7. Multiple Choice — Zero trust

Question

Which statement most appropriately describes zero trust?

A Access from inside is safe, so verification is unnecessary.

B A firewall alone is sufficient.

C No communication is trusted automatically and every access is verified.

D Only outside access needs monitoring.

Answer: **C** — Zero trust verifies every access, including internal access.

8. Short Answer

Question

Give two main uses of a VPN.

Answer: Safe remote connection to an organization's network and safe connection between offices/branches.

Question

Why is a public-facing web server placed in a DMZ?

Answer: To separate it from the internal network so an attack on the public server is less likely to reach internal data.

Question

Why is a firewall alone not enough?

Answer: Other layers can still limit damage if the firewall or another measure is breached.

Question

Why has the traditional perimeter model become less sufficient?

Answer: Cloud services and remote work have blurred the boundary between inside and outside.

EXAM-STYLE QUESTION — [6 MARKS]

Question

Explain why defense in depth protects an organization better than relying on a firewall alone.

Answer: 1. A firewall controls communication entering or leaving the network.
2. A VPN protects communication paths, especially remote connections.
3. A DMZ separates public-facing servers from the internal network.
4. Endpoint protection such as antivirus and updates protects individual devices.
5. If one measure is breached, another layer can still limit damage.
6. Therefore, layered security is stronger than a single barrier.

9. Think as an Engineer — Investigate & Decide

Question

A company has a public website, staff working from home, and confidential customer data.

Recommend a layered design and give two reasons it is safer than a firewall alone.

Answer: Use a firewall at network entry, a VPN for remote staff, a DMZ for the public website, and endpoint protection for devices. Zero trust should verify access. Different layers protect different paths, and if one layer fails the others can still limit damage.

10. New Context

Question

A hospital stores patient records in a cloud service and allows doctors to access them from home.

Using zero trust and defense in depth, suggest two measures, explain where each helps, and identify one remaining risk.

Answer: Require identity/permission verification for every access and use encrypted remote communication such as a VPN. Endpoint protection can also protect devices. A remaining risk could be a compromised account or device.

2-3 — Incident Response and Risk Management

1. Mark each statement (A–D) with ■ if true or × if false.

Question

A. A security incident is a situation in which actual damage occurs because of an information security problem. (■ True / × False)

Answer: True — This is the chapter's definition.

Question

B. Containment is performed before eradication so the spread of damage can be limited. (■ True / × False)

Answer: True — Containment limits the scope before the cause is removed.

Question

C. Risk size can be determined by impact alone. (■ True / × False)

Answer: False — Risk size = impact × likelihood.

Question

D. Prevention of recurrence is the final stage of the six-stage response. (■ True / × False)

Answer: True — The final stage analyzes the cause and prevents recurrence.

2. Answer the following questions.

Question

1. What is a security incident?

Answer: A situation in which actual damage occurs due to an information security problem.

Question

2. Name the six stages of incident response in order.

Answer: Preparation → Detection → Containment → Eradication → Recovery → Prevention of recurrence.

Question

3. What two factors determine risk size?

Answer: Impact and likelihood.

Question

4. What does impact mean?

Answer: How much damage occurs if the risk actually happens.

Question

5. What does likelihood mean?

Answer: How probable it is that the risk will actually happen.

3. Matching — incident-response stages

Question

Match each description with the most appropriate choice.

Choices: A Preparation • B Detection • C Containment • D Eradication • E Recovery • F Prevention of recurrence

a Establish response procedures in advance.

b Discover and confirm that an incident has occurred.

- c Limit the scope of impact so damage does not spread.
- d Remove the cause, such as malware.
- e Return systems and services to normal.
- f Analyze the cause and prevent recurrence.

Answer: $a \rightarrow A \bullet b \rightarrow B \bullet c \rightarrow C \bullet d \rightarrow D \bullet e \rightarrow E \bullet f \rightarrow F$

4. Multiple Choice — security incident

Question

Which is NOT an appropriate example of a security incident?

- A Data is encrypted by a ransomware infection.
- B Customer information is leaked through unauthorized access.
- C A normal software update is performed to add new features.
- D A web service becomes unavailable because of an attack.

Answer: C — A normal software update is not a security incident.

5. Multiple Choice — risk assessment

Question

Which statement best describes risk assessment?

- A Only large-impact risks need attention.
- B Only high-likelihood risks need attention.
- C Risks with both high impact and high likelihood should be addressed first.
- D All risks must be addressed simultaneously.

Answer: C — Risk size is impact $\times$ likelihood, so high-high risks have the highest priority.

6. Fill in the blanks

Question

A situation in which actual damage occurs because of an information security problem is called (a). When it occurs, (b), which limits the spread of damage, is performed before (c), which removes the cause. After that, (d) returns systems to normal, followed by (e), which prevents recurrence.

Answer: (a) security incident • (b) containment • (c) eradication • (d) recovery • (e) prevention of recurrence.

7. Short Answer

Question

Give three examples of security incidents.

Answer: Information leak, data tampering, and service outage.

Question

Why must containment come before eradication?

Answer: Containment limits the spread of damage before the cause is removed.

Question

Why should recovery not be performed before the cause is removed?

Answer: If the cause, such as malware, remains, the same incident may happen again.

Question

What is the risk formula?

Answer: Risk size = Impact $\times$ Likelihood.

Question

Which risks should be addressed first?

Answer: Risks with both high impact and high likelihood.

EXAM-STYLE QUESTION — [6 MARKS]

Question

Explain why the six stages of incident response should be carried out in order, using eradication and recovery as examples.

Answer: 1. Preparation and detection make the organization ready and confirm the incident.
2. Containment limits the spread of damage.
3. Eradication removes the cause, such as malware.
4. Recovery returns systems and services to normal.
5. Recovering before eradication can leave the cause in the system.
6. Therefore, the ordered stages help limit damage and prevent the same incident from returning.

9. Think as an Engineer — Investigate & Decide

Question

A school's student-record system is hit by ransomware: files are encrypted and staff cannot log in. The incident has just been detected. What stages should be carried out next?

Answer: After detection, carry out containment to limit spread, eradication to remove the ransomware/cause, recovery to restore clean systems/data, and prevention of recurrence to analyze the cause and prevent another attack.

10. Risk-prioritization scenario

Question

A school faces two risks: another ransomware attack and a brief power cut. How should impact and likelihood be used to decide which to address first?

Answer: Estimate the impact and likelihood of each risk. The risk with the higher combination should receive higher priority. A high-impact, high-likelihood ransomware risk should be addressed before a lower risk.

11. New Context

Question

An online shop has detected that customer data has been leaked. Using the six-stage response and risk assessment, name the next two stages and explain why. Then identify one risk to prioritize afterward.

Answer: After detection, containment should limit further leakage, followed by eradication to remove the cause. Then assess remaining risks using impact × likelihood and prioritize a high-impact, high-likelihood risk.

12. Reflect & Challenge

Question

Which stage of incident response might be hardest under pressure? Give a reason.

Answer: Answers may vary. For example, containment can be difficult because decisions must be made quickly to stop damage spreading while the full cause may not yet be known.

Question

Suggest one thing an organization can prepare in advance to make incident response faster.

Answer: Examples: a documented response procedure, assigned responsibilities, communication contacts, backup arrangements, or predefined isolation procedures.

Chapter 2 — Mixed Revision

A. Quick-fire questions

Question

What is HTTPS?

Answer: HTTP with TLS encryption added; it protects communication through encryption, tamper detection, and verification of the communication partner.

Question

What is the main role of public-key cryptography in HTTPS?

Answer: Safe key sharing.

Question

What is the main role of common-key cryptography?

Answer: Fast data communication after the common key is shared.

Question

What does MFA require?

Answer: Two or more different authentication factors.

Question

What does a firewall do?

Answer: Allows or denies communication at a network entry/exit point according to rules.

Question

What does a VPN provide?

Answer: An encrypted virtual private line/tunnel over the Internet.

Question

What does a DMZ separate?

Answer: Public-facing servers from the internal network.

Question

What is defense in depth?

Answer: Layering multiple security measures.

Question

What is zero trust?

Answer: No communication is trusted automatically; every access is verified.

Question

What is a security incident?

Answer: A situation in which actual damage occurs due to an information security problem.

Question

What is containment?

Answer: Limiting the scope of impact so damage does not spread.

Question

What is eradication?

Answer: Removing the cause of the incident.

Question

What is recovery?

Answer: Returning systems and services to their normal state.

Question

What is risk size?

Answer: Impact × likelihood.

B. Final Multiple Choice

Question

Which pair is correctly matched?

A Digital certificate — verifies the communication partner

B VPN — removes malware from an endpoint

C DMZ — exposes all servers directly

D MFA — uses only one factor

Answer: A — A digital certificate verifies the communication partner.

Question

Which sequence is correct?

A Detection → Preparation → Recovery → Containment

B Preparation → Detection → Containment → Eradication → Recovery → Prevention of recurrence

C Preparation → Recovery → Detection → Eradication

D Containment → Recovery → Detection → Prevention

Answer: B — This is the six-stage order given in Chapter 2.

Question

Which combination best represents defense in depth?

A Firewall only

B Password only

C Firewall + VPN + DMZ + endpoint protection

D DMZ only

Answer: C — Defense in depth uses multiple layers.

EXAM-STYLE QUESTION — [6 MARKS]

Question

A company wants to protect an online service. Explain how cryptographic technologies, authentication, network security design, incident response, and risk assessment can work together.

Answer: 1. HTTPS can use public-key cryptography for safe key sharing and common-key cryptography for fast data communication.

2. Digital certificates can verify the genuine communication partner and MFA can reduce unauthorized login risk.

3. Firewall, VPN, DMZ, and endpoint protection provide defense in depth.

4. Zero trust verifies access instead of automatically trusting it.

5. If an incident occurs, follow preparation, detection, containment, eradication, recovery, and prevention of recurrence.

6. Prioritize risks using impact × likelihood, addressing high-impact and high-likelihood risks first.